

Kathmandu University School of Education
Department of STEAM Education

Bachelor of Technical Education (B Tech Ed) Program
Information Technology (IT)

In Association with School of Engineering, KU

Course Curriculum

Semester II

2022

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Overview of the B Tech Ed Program

Technical and Vocational Education and Training (TVET) has been one of the prominent sectors globally for a long. Dewey (1916) saw TVET as a tool for education reform in modernizing society. This has been equally vibrant in the present context as scholars (e.g., Maclean & Wilson, 2009) argue that TVET is an education and training that prepares an individual for gainful employment. TVET programs equip young people with the skills, knowledge, and competencies required to enter a particular type of professional career (OECD 2017). Hence, the role of TVET in preparing skilled human resources and enabling them to transition into the career path for gainful employment in specific and the country's economic development in a broad is inevitable. The formal TVET began in Nepal only around mid of the 20th century. Nevertheless, occupational skills have been transformed from generation to generation for a long (Ministry of Education [MoE], 2012). The establishment of the Council for Technical Education and Vocational Training (CTEVT), the governing body of TVET in 1989, was a substantive effort toward its development and expansion (CTEVT, 2019). Additionally, at present, the Centre for Education and Human Resource Development (CEHRD) offering technical diplomas in 284 technical stream schools throughout the country (GoN, 2018) and universities (such as Kathmandu University, Manmohan Technical University, etc.) run bachelor and master in TVET and other 11 federal ministries offer TVET related formal and non-formal programs (MoEST, 2019). The CTEVT, which is in charge of formulating TVET-relevant policies, developing standards for programs and curriculums, coordination, accreditation, monitoring, and supervision (CTEVT, 2019) largely shares the TVET related activities. There are 1131 CTEVT affiliated and constituent institutions with roughly the average capacity of 70,000 per annual operate formal TVET programs (CTEVT, 2020). Further expansion of TVET institutions in the country is yet to be expected soon since the government has a provision of establishing at least one technical school in all local levels.

Bachelor of Technical Education is a program to develop instructors and teachers in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the ‘practical focus’ of the program, the assessment has been proposed to address the needs of experiential learning.

The 136-credit program has been structured four types of the courses, such as Educational Core (18), Vocational Specialization (80), Vocational Pedagogy (20), and Educational Research and Internship (18) with a view to inculcate both pedagogical and vocational skills. The vocational specializations have been offered in a manner that they can be utilize hands-on activities as well. The program begins with the foundational courses in general education, vocational pedagogy and the area of specialization. In the later part of the program, OJT/WBL and forms of experiential learning shall be introduced. The culmination of the program will be internships in their specializations and a research report that addresses the problem of vocational education.

B Tech Ed in Information Technology

Bachelor of Technical Education is a 4-year (8 semesters) program to develop instructors, teachers, professionals in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the ‘practical focus’ of the program, the assessment has been proposed to address the needs of experiential learning. The B Tech Ed in IT program aims at providing technical, vocational, and educational exposures for students who can become IT professionals, IT Instructors, TVET educators, TVET curriculum developers, etc. The occupational/specialization courses emphasize the development of core IT skills in the field of computer science and technology innovation. The B Tech Ed in IT graduates will be eligible for doing master's program in IT, TVET, Education, Social Sciences, etc.

Major Objectives of the B Tech Ed in IT Program

- Demonstrate Level 6 competencies envisaged by the National Vocational Qualification Framework (NVQF) of Nepal,
- Exhibit comprehensive, meaningful, and coherent knowledge and skills in the field of technology and computer science
- Apply occupational specialization knowledge and skills in solving problems occurred in their contexts of work,
- Apply vocational pedagogical knowledge and skills in technical and vocational instruction of Information Technology,
- Show scholarly literacy, communication, quantitative reasoning, critical thinking, learning skills needed for advanced and work-based learning,
- Exemplify a deep and principled understanding of the technical and vocational learning processes and the role of the instructors in facilitating these processes in the students,
- Apply a wide range of teaching process skills including curriculum development, lesson planning, materials development, assessment, and pedagogical approaches, and
- Enable students to become resilient and lifelong learners by harnessing internal and external assets as TVET professionals.

Career Prospects

- TVET Instructors and Educators
- TVET Curriculum Developers
- IT Professionals (Software and Apps developer, programmer, Network Engineer, Web Developer, IT Admin, IT Engineer, etc.)
- Eligible for doing master's program in IT, TVET, Education, Social Sciences, etc.

Requirements

The minimum prerequisite for this program is the completion of Level 5 education (as per National Vocational Qualification Framework) in relevant vocational specializations. The details are-

- Minimum D+ grade in each subject of grade 12 with CGPA 1.8 or more
Or
- A minimum score of second division marks in 10+2, PCL or equivalent in any disciplines

Core Courses

ENGT 100 Communication English

Course: Communication English

Code: ENGT 100

Credit Hours: 3 (2 Th + 1 PR)

Nature: Theory and Practical

Course Description and Goals

This course presents an overview of the communication skills crucial for technical educators and practitioners. It integrates the aspects of technical communication, professional presentations, organizational writing, research orientation, and application of different rhetorical strategies. Primarily, the course aims to enable students to use the English language in technical, pedagogical, and business settings, and to equip them with the skills of analyzing oral and written texts.

Learning Outcomes

On completion of this course, the students will be able to:

- Produce texts effective for technical, educational, and organizational settings
- Construct discourses applying various rhetorical strategies
- Make effective oral presentations in onsite and/or online platforms

Content with Specific Objectives

Specific Objectives	Contents
Use advanced English grammar in writing and speech	Unit One: Advanced Grammar (9 hrs.) • Mixed Tenses; Active and Passive Voice; Reported Speech; Subject-Verb Agreement; Relative Clauses; Connectives/Conjunctions; Conditionals; Inversion and Cleft Sentences; Comparisons; Substitution and Elimination; Prepositions; Articles
Internalize the fundamentals of technical communication	Unit Two: Technical Communication (8 hrs.) • Basics of communication: Communication process; Overcoming communication barriers; Audience recognition • Objectives: Clarity; Conciseness; Accuracy; Organization; Ethics • Writing process: Prewriting; Writing; Rewriting • Basic digital skills: Document design; Web 2.0 skills

Design and deliver effective oral presentation	Unit Three: Oral skills (8 hrs.) Technical presentations: the process, delivery, and update; Preparing visuals; Seminars: (paper-based) and virtual; Workshops and panel discussions; Negotiation skills for professionals; Idea pitching; Classroom communication
Prepare routine texts for business/organizational settings	Unit Four: Organizational /Business Writing (10 hrs.) Memos; Letters: job application, cover, inquiry, recommendations/ references; Emails; Curriculum vitae and Resumés; Team communication: meeting and minutes; Proposals: concept papers, research and project proposals; Reports: conference papers, technical articles and project reports
i) Internalize different types of text organization. ii) Develop critical reading and writing skills	Unit Five: Rhetorical strategies (definition, purpose, audience and strategies) (10 hrs.) a. Description: Objective: spatial/physical, graphic; Subjective Key Reading: Virginia Woolf: “The Death of the Moth” b. Narration: chronology; reverse chronology; flashback Key Reading: Nicholas Gage: “The Teacher Who Changed My Life” c. Process Analysis: How equipment/systems work; How humans perform tasks Key Reading: Armand Dennis: “The Four-Tusked Elephant” d. Comparison and Contrast: point-by-point method; block method Key Reading: Robert Jastrow: “Brains and Computers” e. Cause and Effect: proximate and distant cause/effect Norman Cousins: “Pain Is Not the Ultimate Enemy” f. Persuasion and Argument: AIDA Model (Attention, Interest, Desire, Action); Toulmin’s Model (claim, ground, warrant, backing, rebuttal, modifier) H. L. Mencken: “The Penalty of Death”

Major Teaching and Learning Strategy

- i) Lectures
- ii) Group discussions
- iii) Student presentations
- iv) Team projects

Assessment Plan

- i) In-semester (journal/portfolio, assignments, tests): 50 marks
- ii) End-semester (Objective and Subjective): 50 marks

Books and Reference Materials

- Alley, M. (2013). *The craft of scientific presentations*. 2nd ed. Springer.
- Azar, Betty, Schramper; Hagen, and Stacy A. *Understanding and Using English Grammar*. Pearson Education, 2017.
- Bouvée, C., and Thill, J. V. (2018). *Business communication today*. 14th ed. Pearson.
- Farlex International. *Complete English Grammar Rules*. Farlex International, 2016.
- Gerson, S. J, and Gerson, S. M. (2017). *Technical communication: Process and product*. 9th ed. Pearson Education.
- Trimmer, J, and Hairston, M. (1987). *The riverside reader*. 2nd ed. Houghton Mifflin Company. (Selected chapters for Unit 5)

Vocational Pedagogical Course

VPED 101 Instructional System Design

Course: Instructional System Design

Code: VPED 101

Credit Hours: 2 (1 TH + 1 PR)

Nature: Theory and Practical

Course Description

This course focuses on systems approaches to the design of effective, efficient and appealing instructions, based on learning psychology and instructional strategies. It provides students with the fundamentals of instructional design including the principles of learning theory, and instructional strategies that are relevant to instructional design. This course aims to develop facilitating skills, resource management of training, and development of instructions. The course enables undergraduate students to deliver classroom instructions or training effectively by applying different ways of facilitation skills and methods.

Major Learning Outcomes

In this course, students will be able to:

- aware of multifarious instructional system design theories and its application.
- explore of instructional design and instructional system design model
- Design and develop instructions as per ID models

Contents in detail with Specific Objectives

- Module 1: Introduction to instructional design
- Module 2: Instructional design models and their implications
- Module 3: Planning and executing session
- Module 4: Develop the proper ID for training

Modules	Course Details
Unit 1: Introduction to instructional design (6 hours)	In this module, students will: • know about ID models • historical approaches of ID models • application of ID models • types of ID models • role of ID models in teaching and training
Unit 2: Instructional design models and their implications (10 hours)	• explore motivational theory • design and develop a motivational session • explore ADDIE model • design and develop ADDIE based instructions

	• explore ASSURE model • design and develop ASSURE-based sessions • explore Merrill's First Principles of Instruction • design and develop Merrill's-based sessions
Unit 3: Planning and executing session (8 hours)	In this module, students will: • Gagne's Nine Events of Instruction • design and develop the lesson plan • develop the training package • explore TPACK model • explore SMAR model
Unit 4: Develop the proper ID for training (6 hours)	In this module, students will: • develop and validate the ID models • integrate ID models • application of Kirkpatrick model in an educational evaluation • design and develop Self Dependent Learning Materials

Major Teaching and Learning Strategy

- v) Lectures
- vi) Group discussions
- vii) Teacher and Student presentations
- viii) Team projects

Assessment Plan

- iii) **In-semester:** 50 marks (Assignments, Attendance, Projects, Presentations, Groupwork, Practical in the field, classroom participation, etc.)
- iv) **End-semester** (Objective and Subjective): 50 marks

Books and Reference Materials

Occupational/Specialization Course

ITEX 104 Database System

Course: Database System

Code: ITEX 104

Credit Hours: 3 (2 TH + 1 PR)

Nature: Theory and Practical

Course Description and Goals

This course covers the fundamental concepts about database management system. The main objective of this course is to make the learners familiar with different database systems to be dealt with in real world scenarios. The course begins from the historical development of database systems to its implementation in relational system and conclude with knowledge on crash recoveries. It aims at making the learners familiar for using various database systems, and database programming using SQL. Learners will be at least able to design a database system required at their working place.

Learning Outcomes

On completion of this course, the students will be able to:

- a) understand the basic concepts of database management system.
- b) design a simple database system and normalize it
- c) use SQL programming language to design queries
- d) understand and plan recovery mechanism in case of failure

Content with Specific Objectives

Specific objectives	Contents
1. understand the idea of DBMS 2. gather idea of different databases in modern market 3. understand the development of database systems 4. understand the concept of abstraction, independence 5. understand the idea of schema and instances and implement in real life example	Unit One: Introduction to DBMS (4 Hrs.) 1. Concepts of DBMS 2. Application of DBMS 3. Historical development of database systems (File system, Hierarchical, Network, Relational) 4. Database Architecture 5. Data Abstraction 6. Data Independence 7. Schema and Instances
1. Understand the concept of entity-relationship model	Unit Two: Data Modelling (6 hrs.) 1. Logical, Physical and Conceptual 2. E-R model

2. implement the ideas of entities, keys, attributes in real life examples 3. understand constraints and implement in real example 4. to be able to design ER diagrams in real world scenario	3. Entities, keys, attributes, relationship 4. Constraints 5. E-R diagram practices
1. Understand the concept and use of SQL 2. categorize DDL, DML, DCL and implement with examples 3. implement aggregate functions 4. understand and implement nested subqueries 5. understand the concept of different types of views	Unit Three: SQL (7 hrs.) 1. Introduction to SQL 2. Basic structure of queries 3. DDL, DML, DCL 4. Set Operations 5. Aggregate Functions – GROUP BY, HAVING etc. 6. Nested Subqueries 7. Relations (Joined, derived) 8. Views
1. Understand the concept of relational algebra 2. understand the concept of relational calculus 3. understand various operations and functions in relational algebra 4. understand various activities withing the relational calculus domain	Unit Four: Relational Algebra and Calculus (7 hrs.) 1. Introduction to Relational Algebra 2. Selection and Projection 3. Set Operations 4. Renaming 5. Joins, Division 6. Introduction to Relational Calculus 7. Tuple Relational Calculus 8. Domain Relational Calculus 9. Calculus vs Algebra
1. understand the concept of normalization 2. understand the concept of decomposition	Unit Five: Normalization (6 hrs.) 1. Introduction to Normalization, importance 2. Non loss decomposition 3. Functional Dependencies 4. 1NF, 2NF, 3NF, BCNF, DKNF, 5. Features of a good relational database design

3. understand and implement the concept of functional dependencies 4. normalize to different extent for real world scenarios 1NF, 2NF, 3NF, BCNF, etc	
1. understand the concept and importance of hashing and indexing 2. understand various types of hashing techniques 3. understand various types of indexing techniques	Unit Six: Hashing and Indexing (5 hrs.) 1. Basic Concepts 2. Ordered Indices 3. B ⁺ Tree, B- Tree Indexing 4. Static Hashing, Dynamic Hashing 5. Comparison of Ordered Indexing and Hashing
1. understand the concept of transaction 2. define and understand ACID properties 3. understand serializability 4. understand the concept of deadlock and its prevention 5. understand lock-based and timestamp-based protocols	Unit Seven: Transaction Management and Concurrency Control (5 hrs.) 1. ACID properties 2. Concurrent Executions 3. Serializability, recoverability 4. Lock-based Protocols (2PL, Deadlocks) 5. Timestamp Based Protocols
1. understand the concept of crashing 2. classify different types of failures 3. understand various process of recovery	Unit Eight: Crash Recovery (5 hrs.) 1. Failure Classification 2. Recovery and Atomicity 3. Log-based Recovery 4. Advance Recovery System, Remote Backup systems

Major Teaching and Learning Strategy

The course is designed to be implemented both theoretically and practically. Learners and facilitators both should be involved in classroom activities. Practical demonstration of SQL queries should be done, and students' knowledge should be assessed on timely

basis. Group discussions and participation in group is recommended mostly for depth understanding of the subject matter. Facilitator is responsible for developing classroom materials required for teaching-learning process, use of multimedia is recommended as visuals/images would prove more fruitful for student involvements. Facilitator should at least demonstrate the use of MySQL/ MongoDB or any other database that are most popular in the software market.

Assessment Plan

- a. In-semester:** 50%
- b. End-semester:** 50%

References

1. A. K. Majumdar and P. Bhattacharaya, "Database Management Systems", Tata McGraw Hill, India
2. Elmasri and Navathe, "Fundamentals of Database Systems", Pearson Education
3. "Database Systems: Design, Implementation, and Management", 10th Edition by Thomas Connolly and Carolyn Begg

ITEX 111 Digital Logic

Course: Digital Logic

Course Code: ITEX 111

Credit Hours: 3 (2 TH + 1 PR)

Nature: Theory and Practical

Course Description and Goals

The course covers the basics of digital logic, its implementation and applications. The course includes the fundamental concepts of Boolean algebra, its implementation through Logic Gates, its application for circuit analysis, flip-flops, counters logic devices and synchronous and asynchronous sequential logic circuit and more. The main objective of this course is to introduce the learners with the basic ideas about digital design, its components and implementation.

Learning Outcomes

On completion of this course, the students will be able to:

- e) understand the basics and principles of digital circuitry
- f) identify and understand various digital components and hardware in circuit design
- g) explain the implementation and functions of various components like registers, flip flops, counters
- h) explore the design and development of other complex circuits by self-study

Content with Specific Objectives

Specific objectives	Contents
1. Explain the basic difference between digital and analog quantities 2. Show how voltage levels are used to represent digital quantities 3. Understand various parameters of a pulse waveform like rise time, fall time, pulse width, frequency, period, duty cycle 4. Understand number system concepts, use and conversion	Unit One: Introduction (6 hrs.) 1. Digital Signals, Digital Waveforms 2. Digital Logic, Digital Operations 3. Digital IC, Computers 4. Clock Waveforms 5. Number Systems and conversions 6. Binary arithmetic 7. Different types of Coding (ASCII, BCD, Gray code, Excess 3 Code)

5. determine 1's and 2's complement of a binary number 6. apply arithmetic operations to binary numbers 7. Understand various coding ideas	
1. Explain the basic logic functions of NOT, OR and AND 2. Describe several types of logic operations and algebra 3. Understand Universal gates and special gates 4. Realize the universal gates as the basic gates 5. state and prove de-Morgan's theorem 6. define canonical and standard form of Boolean expression 7. Understand and convert SOP to POS and vice-versa 8. Simplify the Boolean expressions using K-map method for both SOP and SOP form and including Don't Care conditions	Unit Two: Logic Gates (5 hrs.) 9. Basic Gates: NOT, OR, AND 10. Universal Gates: NOR, NAND 11. Special Gates: XOR, XNOR 12. Realization of other Gates from Universal Gates 13. Boolean Algebra 14. K Map, SOP, POS, Don't Care Conditions
1. Explain combinational circuits with their features 2. implement digital logic for half adder, full adder, half subtractor, full subtractor with their functional expressions,	Unit Three: Combinational Logic Design (10 hrs.) 1. Multiplexer, Demultiplexer 2. Encoder, Decoder 3. Half Adder, Full Adder 4. Parity Generator, Checker 5. Concept of ROM, PROM, EPROM, PLA, PAL

logic diagram and truth tables and timing diagram 3. explain the concept of encoder and decoders 4. design a logic circuit to decode any combination of bits 5. Use BCT to 7 segment decoder in display systems 6. Describe decimal to binary priority encoder, 7. implement 4-bit parallel adder 8. explain functioning of 9-bit parity generator and checker 9. Explain the concept of programming logic with reference to PROM, EPROM, PAL and PLA with circuit and tables	
1. Differentiate between latch and flip-flop 2. Differentiate between level triggering and edge triggering with their features 3. Explain RS, JK, JK master-slave, D&T flip-flops with their logic diagram, graphical symbol, characteristics table, excitation table 4. How SISO, SIPO, PISO and PIPO shift registers operate 5. Explain ripple counter with circuit, state, and timing diagram	Unit Four: Counter and Registers (13 hrs.) 1. Flip flops: RS, JK, JK master-slave, D & T flip flops (level trigger, edge trigger, excitation table) 2. Synchronous and Asynchronous Counters 3. Ripple Counter 4. Ring Counter 5. Modulus Counter 6. Decade Counter 7. A digital Clock 8. Types of Registers 9. Serial in Parallel Out 10. Serial in Serial Out 11. Parallel in Parallel Out 12. Parallel in Series Out 13. Shift Registers

6. explain ring counter with circuit, state, and timing diagram 7. Explain modulus counter with circuit and state diagram	
1. define finite state machine with examples 2. explain Mealy and Moore models of finite state machines 3. describe state, state diagram and state table of sequential circuit 4. understand and describe the design procedure of sequential machines 5. use the flip flops to realize the synchronous machines 6. understand the basics of asynchronous sequential circuits	Unit Five: Sequential Circuits and design (7 hrs.) 1. Concept of state, state diagram 2. Transition Tables, redundant states 3. Using flip flops to realize the synchronous models 4. Asynchronous Sequential Circuits concepts and design
1. to understand the basic working of a processor 2. describe the functions and working of ALU 3. to describe bus architecture 4. to design a basic ALU using logic gates and simple components.	Unit Six: Introduction to ALU Design (4 hrs.) 1. CPU overview 2. ALU 3. Bus architecture 4. Designing a basic ALU

Major Teaching and Learning Strategy

The facilitator and the learners will both develop their level of understanding and information about the subject matter. The digital design components should be explored as practically as possible. Learners need to be more enthusiastic to learn the theories and participate in its implementation to design digital circuits to their best

knowledge. Group discussions and participation in group is recommended mostly for depth understanding of the subject matter. Facilitator is responsible for developing classroom materials required for teaching-learning process, use of multimedia is recommended as visuals/images would prove more fruitful for student involvements.

Assessment Plan

- a. In-semester:** 50%
- b. End-semester:** 50%

References

- a. Brain Holdsworth, “Digital Logic Design”, Elsevier Science, Latest Edition
- b. M. Morris Mano, “Digital Logic & Computer Design”
- c. Donald P. Leach, Albert Paul Malvino and Goutam Saha, “Digital Principles and Applications” Tata McGraw-Hill

ITEX 103 Object Oriented Programming

Course Title: Object Oriented Programming

Code: ITEX 103

Credit Hours: 3 (2 TH + 1 PR)

Nature: Theory + Practical

Course Description and Goals

The course covers the concepts of Object-Oriented Programming using C++ programming language. The main objective of this course is to deliver and to make learners understand the basic concepts of OOP, release the peculiarity of Object-Oriented Paradigm and implement these concepts in real world projects. It covers the OOP concepts like objects, classes, polymorphism, overloading, inheritance, exceptions, file handling and more.

Learning Outcomes

On completion of this course, the students will be able to:

- i) understand the specialty and importance of Object-Oriented Approach
- j) implement the programming features of OOP
- k) create programs and projects using C++ programming language
- l) learn any other OOP language based upon the basics delivered through this course

Content with Specific Objectives

Specific objectives	Contents
1. Compare procedure and object oriented programming concept 2. Describe the feature of object-oriented programming. 3. List different types of OOPs 4. Compare coding structure of C and C++. 5. Understand the details of C++ program structure. 6. Revise the concepts of control structures in C++	Unit One: Introduction to OOP (5 hrs.) 8. Software Crisis and emergence of OOP 9. Procedural vs OOP Languages 10. Features of OOPs 11. Programming structures in C++ 12. Basic programming terminologies: tokens, data types, operators etc. 13. Programming Construct Control Structures
1. understand the basics of classes and objects	Unit Two: Object and Classes (8 hrs.) 15. Introduction to Objects and Classes

2. be able to initialize objects in accordance to its class 3. implement the use of this operator 4. understand the idea of access specifiers 5. understand and implement the use of constructor and destructors in C++	16. Defining a class with data members and member functions 17. Access specifiers 18. Initializing objects 19. Static Data Members 20. Static Member Functions 21. this operator 22. Constructor 23. Destructor
1. To understand and demonstrate the use of operator overloading in C++ 2. implement unary operator overloading 3. implement binary operator overloading 4. implement the concept of operator overloading in member functions and non-member functions	Unit Three: Operator Overloading (7 hrs.) 1. Introduction to operator overloading 2. Overloadable operators 3. Syntax and rules of operator overloading 4. Unary operator overloading 5. Binary Operator overloading 6. Operator overloading with Member and non-member functions 7. Overloading binary operator using a friend function
1. understand and implement the concept of inheritance 2. understand base class and derived classes 3. implement constructors and destructors for derived class in C++ 4. understand concept of abstract class and virtual base class 5. implement use of access specifiers in C++ programs	Unit Four: Inheritance (7 hrs.) 1. Introduction to inheritance 2. Base class and derived class 3. Derived class constructor 4. Derived class destructor 5. Single inheritance and multiple inheritance 6. Virtual Base Classes and Abstract Classes 7. Public, Private and Protected access and control
1. Understand the concept of polymorphism	Unit Five: Polymorphism (9 hrs.) 1. Introduction to polymorphism

2. implement new and delete operators 3. understand and implement virtual functions in C++ programs 4. understand the concept of dynamic binding	2. New and Delete operator 3. Pointer to objects 4. Pointer to derived class 5. Virtual Functions 6. Friend Functions 7. dynamic cast operator 8. typed operator
1. Understand the idea of templates and frameworks 2. to implement class templates in C++ 3. understand function templates	Unit Six: Templates (4 hrs.) 1. Concept of templates 2. Class templates 3. Function Templates 4. Concept of Frameworks
1. Understand the idea of error and exceptions 2. implement ideas to handle exceptions in C++ 3. implement rethrowing an exception in C++ 4. handle various types of exceptions in C++	Unit seven: Exception Handling (5 hrs.) 1. Error Handling 2. Exception handling constructs (try, catch, throw) 3. Multiple exception handling 4. Catching all exceptions 5. Exception with arguments 6. Re-throwing an exception 7. Handling uncaught and unexpected exceptions

Major Teaching and Learning Strategy

This course is designed to be implemented in both practical and theoretical basis.

Understanding the basics under facilitators guidance and supervision by being involved in classroom activities will be used to complete the theoretical requirements.

Implementation of the theoretical knowledge in the computer room or in personal computers will entertain the practical requirements of the course. Both the learner and the facilitator are expected to progress in developing depth knowledge regarding the subject matter in time to come.

Assessment Plan

a. In-semester:

Students' evaluation will be carried out throughout the semester according to their performance and participation in classroom activities related to the course. The classroom activities may include two or more of any of the following:

1. presentations
2. project works
3. practical
4. group discussions
5. knowledge sharing
6. internal examinations/tests etc.

b. End-semester

References

- c. Deitel and Deitel, C++ How to Program, Latest Edition, Pearson Publication.
- d. Joyce Farrell, Object-oriented programming using C++, Fourth Edition, Cengage Learning
- e. Daya Sagar Baral and Diwakar Baral, “The Secrets of Object-Oriented Programming in C++”, Latest Edition.

ITEX 121 Computer Maintenance and Troubleshooting

Course Title: Computer Maintenance and Troubleshooting

Course Code: ITEX 121

Credit Hours: 2 (1 TH + 1 PR)

Nature: Theory & Practical

Course Descriptions

The course helps learners develop fundamental knowledge and understanding relating to computer hardware, Troubleshooting & General PC Problems, Operating system installation and restore and backup.

Objectives/Learning Outcomes:

- Explore the knowledge and skills in computer hardware
- Develop knowledge and skills the computer microprocessor
- Assemble and dissemble of computer
- Be familiar with computing devices troubles and handling
- Develop knowledge and skills in Operating system installation and restore.
- Develop skills in Safe and smooth operation of computing devices

COURSE	Description	Hours
PC hardware /introduction to Microprocessor	• The PC hardware consisting of input, processing and output sections and basic building components. • Introduction to computer hardware components of mother boards, CPU, chipset, various ports, slots, connectors, addon cards. • Protection of PC hardware, anti-static wrist band, protection and safety devices • Evolution of Microprocessor, architecture of Microprocessor.	2
Primary Memory/ Secondary storage	• ROM, PROM, EPROM, EEPROM, L1, L2 RAM, types of memory, static, dynamic, DRAM, SDRAM, DDR2, DDR3. Virtual memory, Cache memory, Linear & Physical memory, video memory. • HDD like IDE, SATA, e-SATA, SCSI, Introduction to HDD controllers like SCSCI controllers and RAID controllers their requirement and configuration.	2

	Backup devices magnetic tape drives, UBS Pen-drives, External HDDs, CDROM, CDRW, DVD, Blue-Ray Discs.	
Power Supply	Switched Mode Power supply block diagram, working principles, testing and troubleshooting, power rating, requirement of SMPS wattage depending on parameters like processor, HDDs used.	2
Troubleshooting & General PC Problems	- Introduction, General Troubleshooting rules, Preventive Maintenance. Typical Motherboard BIOS, BIOS Features, BIOS & Boot Sequences, BIOS Shortcoming & Compatible Issues, BIOS Troubleshooting, BIOS Upgrades. POST, - Error Code: Beep Code, Post Code, Post Reader Card Basic Memory Concepts: Introduction, Installing Memories, Upgrade Options & Strategies - Printers: Printer Technology, How Printer Works, Attaching Printer, Installing Network Printer Drivers, Common Printer Problems & Solution	5
Operating system Back-up and restore	- system Image backup, backup and restore, freeing up disk space, defragmentation, taking updates, network firewall, spyware and unwanted software protection, - run maintenance, and other operating system security features - Upgrade Options & Strategies - Introduction to FAT/NTFS, difference between FAT/NTFS. data storage and data access principles of FAT/NTFS, FAT and MFT structure, attributes in FAT/NTFS, file management and memory management in FAT/NTFS, data deletion and data recovery Concept. formatting,	3

Practical-1	Study of different types of motherboards.	1
Practical-2	Study of jumper settings on mother boards.	1
Practical-3	Installation of memory modules.	1
Practical-4	Study of Various adapter cards and their functioning and installation.	1
Practical-5	Study of different buses and the number of pins in the different slots corresponding to different buses.	1
Practical-6	Opening the PC and identification and study of its different blocks, disassembling.	1
Practical-7	Study of various units, types of display Graphic cards.	1
Practical-8	Assembly and disassembly of different Desktop /Laptop	1
Practical-9	Identification of all chips and crystals on laptop and desktop motherboard	1
Practical -10	Installation of DVD/USB.	1
Practical-11	Study of faults diagnosis based on different beeps.	1
Practical-12	Configure CMOS, BIOS setup.	1
Practical-13	Installation of hard disk, Partitioning.	1
Practical-14	Installation of hard disk in master and slave mode.	1
Practical-15	How to access the configured space of ISA slot.	1
Practical-16	Installation of Windows NT Server /Linux, clients and practice of using the network 2 Practical-8. Running of Scan disk and Disk defragmenter as part of preventive maintenance	1
	Total	30

Major Teaching and Learning Strategies

Lecture, Demonstration, PowerPoint Presentation, Problem Solving, Group Discussion, Technology-integrated teaching and learning, Inquiry-based learning

Assessment:

Internal: 50% (Project work, Practical, Data analysis, Demonstration, Groupwork, Presentation, etc.), Practical listed in the syllabus

External: 50%

References

1. Learn the Basics of Linux, David Maxwell
2. Laptop Repair Complete Guide; Including Motherboard and Component Level Repair! Garry Romaneo.
3. Laptop Chip Level Repair Classroom Notes, <http://laptoprepaircourse.in>
4. Computer Basics: Basic Troubleshooting Techniques (gcfglobal.org)

ITEX 120 Probability and Statistical Modeling

Course Title: Probability and Statistical Modeling

Course Code: ITEX 120

Credit Hours: 3 (2 TH + 1 PR)

Nature: Theory + Practical

Course Description and Goals

This course aims to explore the application of probability and statistics for computer application. Students will use the techniques and strategies of statistical modeling and apply in computing field. The objective of the course is to provide students with a clear understanding of the basic statistical concepts and tools and to enable them to use these tools as Necessary Avenue foreengineering professions and scientific knowledge.

Learning Outcomes

On completion of this course, the students will be able to:

- Relate statistics with computing
- Analyze different statistical models for data analysis
- Use the concept of probability and statistics in modeling mathematical data.

Course Contents

Specific Objectives	Contents
• Interpreting the importance of statistics and data • Present data in various charts and figures (Dot plot, scatter plot, histogram, Stem-and-leaf plot, box plot, pareto chart) and describe their usage • Interpret data numerically by using various parameters of the data (Central tendency and measure of dispersion)	Unit 1: Introduction to Statistics and Data Description (7 hours) • Graphical Presentation of Data i. Dot Plots and Scatter Plots ii. The frequency Distribution and Histogram iii. The Stem-and-leaf Plot iv. The Box Plot v. The Pareto Chart • Numerical Description of Data i. Measures of Central Tendency: Mean, Median, Mode, Mean of combinedgroups, Comparison of mean, median and mode. ii. Measures of Dispersion: Range, Quartile deviation, Standard deviation & Variance, Coefficient of Variation, Skewness and Kurtosis

• Interpret the ideas and importance of probability in mathematics and real world • Describe random experiment, sample space, events, etc.) • Define probability, finite sample space, enumeration, conditional probability and interpret their usage • Describe and apply Bay's theorem	Unit 2: Probability (5 hours) • Introduction • A Review of Sets • Random experiment, Sample space and Events (simple and composites), Mutually exclusive and Collectively exhaustive events, Independent events • Probabilities definition and Assignment • Finite Sample Space and Enumeration • Conditional probability • Partitions, Total probability, and Bayes' theorem and its applications
• Define random variable, distribution function, discrete and continuous variables • Interpret the characteristics of distributions with examples	Unit 3: One Dimensional Random Variables (3 hours) • Introduction • The Distribution Function • Discrete and Continuous Random variable • Some Characteristics of Distributions (mean, variance)
• Describe function of one random variable and mathematical expectation • Provide examples and interpret function of discrete and continuous random variable • Solve problems related to the concepts	Unit 4: Functions of One Random Variable and Mathematical Expectation (3 hours) • Introduction • Equivalent Events • Function of Discrete and Continuous Random variable • Mathematical Expectation.

• Define and describe discrete distributions with examples • Discuss Bernoulli trials and distribution with examples • Define and discuss the binomial distribution with examples • Use binomial distribution to solve the problems • Define Poisson distribution with examples	Unit 5: Some Important Discrete Distributions (4 hours) • Introduction • Bernoulli Trials and the Bernoulli Distribution • The Binomial Distribution i. Mean and variance of Binomial Distribution ii. The cumulative Binomial Distribution iii. An application of Binomial Distribution • The Poisson Distribution i. Mean and variance of Poisson Distribution ii. The Poisson Approximation to Binomial Distribution
• Define and describe normal distribution with real-life examples • Examine the use of normal distribution • Demonstration the place of mean, variance, mode, and their usage in normal distribution • Solve problems related to normal and standard normal distribution • Define and use central limit theorem • Define and discuss the normal approximation to binomial distribution	Unit 6: The Normal Distribution (4 hours) • Introduction • Properties of the Normal Distribution • The Mean and Variance of the Normal Distribution • The Normal Cumulative Distribution • The Standard Normal Distribution • Problem-Solving Procedure • The Central Limit Theorem • The Normal Approximation to Binomial Distribution
• Define and interpret the real-world meanings of sample, population,	Unit 7: Random Samples and Sampling Distributions (4 hours)

sampling, census, estimate, estimator, parameters, and statistics with examples • Discuss random sample with examples • Interpret and use the t-distribution in problem solving • Interpret and use the Chi-Square distribution in problem solving • Interpret and use the F-distribution in problem solving	• Population and sample, Census and sampling, Estimate and estimator, Parameter and statistic • Random Samples • Statistics and Sampling Distributions • The Chi-Square Distribution • The t-Distribution • The F-Distribution
• Describe point estimation and interval estimation with their key differences • Discuss the properties of estimator • Interpret the meaning and usage of single-sample confidence interval estimate in the case of mean and variance • Use and interpret two-sample confidence interval to solve related problems (mean and variance)	Unit 8: Estimation (4 hours) • Point Estimation, Interval estimation • Properties of Estimators • Single-Sample Confidence Interval Estimation (mean and variance) • Two-Sample Confidence Interval Estimation (mean and variance)
• Analyze the use of tests of hypothesis in real world • Use tests of hypothesis on a single sample (mean and variance) • Use test of hypotheses on two samples (mean and variance)	Unit 9: Tests of Hypotheses (4 hours) • Introduction • Tests of Hypotheses on a Single-Sample (mean and variance) • Tests of Hypotheses on two Samples (mean and variance)
• Define simple linear regression with example and interpret the real-world meanings of it • Define correlation with real-world examples	Unit 10: Simple Linear Regression and Correlation (4 hours) • Simple Linear Regression and interpretation • Correlation and interpretation

• Use the formula of correlation and regression to solve very basic problems	• Coefficient of determination
• Define and interpret the meaning of statistical quality • Interpret the usage of control charts for measurements and attributes	Unit 11: Statistical Quality Control (3 hours) • Introduction, Statistical Process Control • Control Charts for Measurements • Control Charts for Individual Measurements • Control Charts for Attributes

Major Teaching and Learning Strategies

Lecture, Demonstration, PowerPoint Presentation, Problem Solving, Group Discussion, Technology-integrated teaching and learning (use of GeoGebra, Excel, SPSS, or other statistical software)

Assessment:

Internal: 50% (Project work, Data analysis, Demonstration, Numerical Assignments, Groupwork, Presentation, etc.)

External: 50%

Textbook:

Hines, W. W., Montgomery D. C., Goldsman, D. M., Borror, C. M. (2003). *Probability and statistics in engineering* (4th ed.). John Wiley and Sons, Inc.